

5. SOLID Principles Mapping

- **Single Responsibility Principle (SRP):** TicketPrinter only formats output. PriceCalculator only computes totals. Booking only tracks transaction state.
- **Open/Closed Principle (OCP):** New payment mechanisms (NetBanking, Crypto, GiftCard) extend Payment without editing BookingService or existing classes.
- **Liskov Substitution Principle (LSP):** All Payment subclasses conform to bool pay(double) without unexpected preconditions or special-case checks.
- **Interface Segregation Principle (ISP):** Payment maintains a focused interface. No artificial methods like refund() or sendOtp() are forced upon concrete classes.
- **Dependency Inversion Principle (DIP):** BookingService::processPayment depends on the abstract Payment base class rather than concrete implementations.